A-A-53791
30 December 1988

COMMERCIAL ITEM DESCRIPTION

TESTER, CURRENT LEAKAGE

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a precision, autoranging meter suitable for conducting measurements of leakage current, voltage and resistance during safety testing of medical equipment.

Salient characteristics.

The tester shall be a precision, portable, self contained, auto ranging meter suitable for conducting measurements for leakage current, voltage and resistance during safety testing of medical electrical equipment. The tester shall consist of a single unit incorporating all required switches, terminals, indicators, and meters in a single chassis.

The tester shall be capable of operating on 115 volts and 230 volts, 50/60 Hz power (a selectable switch or changing power cord connections in accordance with supplied instructions is acceptable, a conversion kit is not acceptable). The unit shall be supplied in the 115 volts, 60 Hz configuration for U.S. operation. The tester shall be capable of operating at least a 17 amp draw device at 115 volts and at least 6 amp draw at 230 volts. The unit shall be supplied with a hospital grade plug and suitable power cord at least 6 feet long. The unit shall operate as specified at a minimum ambient temperature range of 15 to 35^{0}C (59 to 95^{0}F).

AMSC/NA FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The tester shall be equipped with a 3-1/2 digit, digital readout meter calibrated in the following units and full scale ranges (+ or - 1 most insignificant digit) including at least:

Current: 0 to 199.9 microamperes
0 to 1999 microamperes

Voltage: 0 to 199.9 volts
0 to 500 volts

Resistance: Two terminal, 0 to 19.99, 0 to 199.9 ohms
Four terminal, 0 to 199.9, 0 to 1999 milliohms

The tester shall be equipped with an on-off switch, 1 red and 1 black probe, 1 U.S. 115 volt/60 Hz test receptacle, terminals for simultaneous connections to a five-patient leads plug of an electrocardiograph and connection to the chassis of the ECG. The tester shall be equipped with switches that can independently disconnect electrical ground and reverse electrical polarity at the test receptacle.

The colored probes shall each consist of a wire at least 4 feet long including its ending of a 4 inch long X 1/4 inch (nominal diameter) rigid handle with a conductive tip or alligator clip. The opposite ends of these wires shall either end in a plug for inserting into the proper terminal of the tester or be permanently attached to the connector. The handle, wire, and plug (if provided) shall be suitably insulated and uniformaly colored, yielding one black and one red probe. The plug or tester end of the red probe shall be identified as positive and the corresponding end of black probe shall be identified as negative. The colored probes shall permit the tester to be utilized as a multimeter.

The tester, when connected to an electrocardiograph in accordance with manufacturer's instructions, shall be capable of making each of the following measurements by activating appropriate switches but without changing terminal connections.

a. Line voltage of AC power line at test receptacle, threeways; neutral to hot, neutral to ground, and hot to ground.

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b. Leakage current from the chassis of an electrocardiograph to ground; from each individual patient electrode to ground; and from all patient electrodes in common to ground with the AC power supply to the electrocardiograph normal and reversed, grounded and ungrounded.

c. The tester shall be provided with a separate carrying case or a detachable cover. The size of the unit, less the separate carrying case (if provided) shall not exceed 0.40 cubic feet (6.55 cubic centimeters). The unit weight, less separate carrying case (if provided) shall but exceed 10 pounds (4.54 Kg). The case shall provide room for all necessary accessories including the instrument cord and plug, as well as the test probes.

The unit shall meet all applicable requirements of UL 1244, standard for Electrical and Electronic Measuring and Testing Equipment.

The unit shall be provided with two complete sets of instructions for operation, maintenance and repair. Instructions shall include pictures or illustrations, as necessary, with complete schematics an wiring diagrams. Instructions shall also contain a list of replaceable parts.

Workmanship. The tester shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One tester, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defines as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling. Sampling for inspection shall be conducted in accordance with MIL-STD-105 and as specified. Inspection shall be conducted at Inspection level II with an AQL of 2.5 defects per hundred units. The unit of product for sampling purposes shall be one current.

Tests. Tests shall be conducted to determine compliance with specified requirements. Testing shall include but not be limited to the following:

When connected to an electrocardiograph in accordance with manufacturer's instructions, test the ability to make the measurements of voltage and leakage current as specified.

Using two resistors of known resistance (one between 100 and 150 oh, and another between 1000 and 1500 millohms) measure the resistance of each in accordance with the manufacturer's instructions and verify readings are within specified accuracy.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standard, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6625-01-142-8233	TESTER, CURRENT LEAKAGE, Medical Equipment, 115/230 volt, 50/60 Hz

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-CSS
PHS
FDA-MPQAS

Project No. 6625-0696

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53791	TESTER, CURRENT LEAKAGE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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